Risk Minimized Portfolio Construction through Algorithmic Modeling

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Abstract

Portfolio optimization remains a fundamental challenge in quantitative finance, yet institutional-quality tools remain largely inaccessible to retail investors. This paper presents a comprehensive portfolio management system that democratizes sophisticated optimization strategies while providing educational value. The system combines Hierarchical Risk Parity (HRP) and Modified Markowitz Mean-Variance Optimization in a novel hybrid approach, driven by a 28-question risk assessment that generates truly personalized portfolios rather than categorical allocations. Unlike traditional robo-advisors that map users into discrete risk categories, our approach produces a continuous spectrum of portfolios—theoretically 10^{28} unique combinations—ensuring each user receives allocations tailored to their specific risk capacity, behavioral tolerance, and thematic preferences. Backtesting on 2019-2024 market data demonstrates that the hybrid approach achieves a Sharpe ratio of 0.97 compared to 0.93 for the S&P 500, while reducing maximum drawdown from 33.7% to 26.7%. The complete system, including paper and live trading integration via Alpaca Markets API, is deployed on Google Cloud Run and available as open source, representing a concrete step toward democratizing institutional-grade investment strategies.

1 Introduction

I started investing at the age of 12 when my mother, who had invested in stocks over the previous 25 years, prompted me to get into markets. Although peers at first were skeptical about this foundation since they could not comprehend why a teenager was talking about portfolio theory, it formed my knowledge about long-term wealth creation and why investment knowledge must be democratized.

The current investment environment is a dangerous paradox. Conventionally considered to be a diversified index, the S&P 500 is now frighteningly concentrated. The seven tech giants, the "Magnificent Seven" (Apple, Microsoft, Amazon, Google, Meta, Nvidia, Tesla) have become close to 30% of the total market capitalization of the index [**?**]. This level of concentration has introduced concealed systematic risk to millions of passive investors on the belief that they are diversified appropriately.

The majority of retail investors default to market-cap weighted indices without being aware of their concentration risks. They have no access to institutional-grade instruments such as Hierarchical Risk Parity (HRP) [1, 2] available to give actual diversification of uncorrelated asset classes. Moreover, despite the existence of advanced tools, they usually provide such measures as Sharpe ratios, maximum drawdown, and correlation matrices without any educational background and also without providing the user with a chance to make informed decisions.

This project is a response to these gaps in that it establishes a platform accessible to every individual that not only builds portfolios that are risk-minimized but also is an educational resource. All metrics will be streamlined and presented in the application to convert complicated financial terms into economic realities that enable users to make optimal investment choices.

The proposed system creates a full-fledged portfolio management platform that democratizes institutional-quality strategies while offering educational value. The system evaluates personal risk profiles using a comprehensive 28-question assessment, develops diversified portfolios across multiple ETF asset classes using Hierarchical Risk Parity and Markowitz optimization, focuses on reducing risk rather than maximizing returns, provides an interactive learning platform with simplified metric explanations, and enables both paper and real trading through the Alpaca API.

2 Related Work

In modern financial theory, portfolio optimization is a well-known concept and therefore used in asset the management and quantitative finance industry. These companies try to find the best way to allocate assets while meeting the expected returns and maintaining the risks.

Among all the concepts, mean-variance theory [3] is the main idea behind modern finance introduced by Markowitz. He developed a mathematical model of diversification in a group of assets. This principle was then used by Warren Buffett with a famous saying, "never put all the eggs in one basket." Markowitz showed that, for a given variance or standard deviation, investors can actually make efficient portfolios by maximizing expected returns.

On top of Markowitz's foundational model, Fabozzi, Markowitz, and Gupta [4] introduced more theoretical concepts such as the efficient frontier, indifference curves, and the role of investor utility in identifying optimal portfolios. Their work emphasized that portfolio construction is not just about combining assets but quantifying the relationship between expected return, risk, and correlation. They showed how diversification can reduce overall portfolio variance even when individual assets can be risky. This reinforced the mathematical and behavioral logic behind mean-variance analysis, establishing the groundwork for later developments in computational and algorithmic optimization approaches.

While the model was a ground-breaking advancement in modern finance, the assumptions of normally distributed returns and stable correlations among assets don't hold true in real markets. Later, the concepts were more extended, including conditional value-at-risk (CVaR) [5]. CVaR attempts to give explanation for these shortcomings in actual market by keeping alternative measures of risk. However, the old optimization approaches are computationally hard to manage even in the presence of realistic limitations like transaction costs, cardinality limits or sectoral exposure caps.

In an attempt to overcome such constraints, researchers have moved to heuristic and meta-heuristic optimization algorithms that can handle nonlinearity and thus NP-hard nature of real-world portfolio problems. Global distribution system algorithms based on genetic algorithm (GA), particle swarm optimization (PSO), ant colony optimization (ACO) and differential evolution (DE) [6] have proved to be useful to find nearly optimal answers to the shortcomings of optimization.

A full rundown by Gunjan and Bhattacharyya [7] examined that GA and PSO are reliable in their performance, relative to traditional quadratic programming methods in solving constrained portfolio problems. They stated that swarm-based and evolutionary models, specifically the firefly curves, artificial bee colony, and bat have better adaption to the volatile and dynamic markets. This results in better risk-return trade-offs. The multimodal ability of these methods seems to make them an appealing alternative to classical optimization.

Recently, with the rise of artificial intelligence (AI) and machine learning (ML) techniques, portfolio optimization has been further transformed. With rise of GPUs and advancement of AI models, it has been easy to compute exponentially large amounts of historical and real-time data. This has led to finding narrow and complex patterns within asset behavior. Advanced machine learning techniques, support-vector machines, and reinforcement learning [8] have been used for asset returns prediction, volatility management, and the construction of adaptive portfolios which can learn from the market. It has shifted the paradigm from static theory-based optimization to dynamic data-based decision making systems, adjusting in an evolving market.

Chakrabarty and Biswas [9] proposed Strategic Markowitz Portfolio Optimization (SMPO) to enhance returns while managing risk, demonstrating improved performance over traditional approaches. Similarly, Paleologo [10] gave a complete frameworks for integrating quantitative methods with fundamental analysis, filling the gap between theoretical optimization and practical implementation.

In addition to AI and heuristics, risk-based framework opportunities have emerged such as the concept of hierarchical risk parity (HRP) in an attempt to moderate the concentration of risks of many of the standard market-cap weighted indices. Detailed by Lopez de Prado [1], HRP allocates capital into uncorrelated asset groups based on hierarchical clustering which does away with needing to do unstable covariance-matrix inversions. Interestingly enough, the methodology will actually eliminate portfolio concentration - which is a concern especially in today's markets which is very concentrated around a few large-cap technology giants known as Magnificent Seven. Recent work by Kalytis et al. [2] compared Hierarchical Risk Parity (HRP) against traditional methods, showing superior performance in out-of-sample tests and validating the approach's effectiveness.

Despite these developments, there are major problems still to be solved in the literature. In particular, many AI-driven models are not very interpretable, and act as "black boxes" hiding the logic behind investment choices. Furthermore, institutional quality optimization technology is largely unavailable to retail investors, who still have to buy and hold various assets as a retail investor without cutting-edge technology. Finally, the educational aspect of portfolio construction, that is to say helping users understand and interpret the notions of risk building (Sharpe ratio, volatility, correlation, etc.) has barely been considered. Any new research should seek to apply bridging functions between institutions' quantitative finance and retail investor's risk minimization, transparency, and accessibility. By working on such reform, contributions will be going into the emerging field of democratized AI-led finance, a domain in which optimized level distribution techniques and ML algorithms must be available to users in an easily interpretable human-like format.

3 Data

The foundation of any portfolio optimization system lies in the quality and diversity of its underlying data. For this project, I assembled a comprehensive dataset that spans multiple asset classes, drawing from both market data sources and user-generated behavioral information. The goal was to create a system that not only understands the mathematical relationships between assets but also captures the nuanced preferences and risk tolerances of individual investors.

3.1 ETF Universe Selection

Rather than attempting to optimize across thousands of individual securities, which would introduce unnecessary complexity and potential overfitting, I curated a focused universe of twenty Exchange-Traded Funds (ETFs) that provide broad market exposure across distinct asset categories. This approach aligns with the democratization goal of the project—retail investors benefit more from a well-diversified ETF portfolio than from attempting to pick individual stocks, a practice that even professional fund managers struggle to do consistently.

The stock category includes eight ETFs spanning different investment styles and geographic regions. The bond category comprises five ETFs representing the fixed-income spectrum. Alternative investments round out the universe with three ETFs providing exposure to non-traditional asset classes. Table 1 presents the complete ETF universe.

3.1.1 Risk Level Classification

Each ETF in the universe is assigned a risk level from 1 to 10, reflecting its historical volatility and potential for drawdown. Table 2 summarizes the risk distribution across asset categories.

Short-term government bonds receive the lowest risk rating, while cryptocurrency and high-growth technology funds receive the highest. These risk levels serve as an initial filter—conservative investors will not be allocated to high-risk assets regardless of their potential returns.

3.2 Market Data Sources

Historical price data forms the backbone of the optimization algorithms, enabling the calculation of expected returns, volatilities, and crucially, the correlation structure between assets. I utilize

Table 1: Complete ETF Universe

Type	Ticker	Name	Category	Risk	Expense
Stock	VOO	Vanguard S&P 500 ETF	U.S. Large Cap	6	0.03%
Stock	VTI	Vanguard Total Stock Market ETF	U.S. Total Market	6	0.03%
Stock	QQQ	Invesco QQQ Trust	Technology/Growth	8	0.20%
Stock	SCHD	Schwab U.S. Dividend Equity ETF	Dividend Income	5	0.06%
Stock	VTV	Vanguard Value ETF	U.S. Value	5	0.04%
Stock	VWO	Vanguard FTSE Emerging Markets ETF	Emerging Markets	7	0.08%
Stock	VXUS	Vanguard FTSE All-World ex-US ETF	International	6	0.07%
Stock	XLU	Utilities Select Sector SPDR Fund	Utilities/Defensive	4	0.09%
Bond	BND	Vanguard Total Bond Market ETF	U.S. Aggregate	3	0.03%
Bond	AGG	iShares Core U.S. Aggregate Bond ETF	U.S. Aggregate	3	0.03%
Bond	TLT	iShares 20+ Year Treasury Bond ETF	Long-Term Treasury	5	0.15%
Bond	GOVT	iShares U.S. Treasury Bond ETF	U.S. Treasury	2	0.05%
Bond	TIP	iShares TIPS Bond ETF	Inflation-Protected	3	0.19%
Alt	GLD	SPDR Gold Shares	Precious Metals	5	0.40%
Alt	VNQ	Vanguard Real Estate ETF	Real Estate/REITs	6	0.12%
Alt	BITO	ProShares Bitcoin Strategy ETF	Cryptocurrency	10	0.95%

Table 2: Risk Level Summary by Asset Category

Asset Category	Risk Range	Avg. Volatility	Role in Portfolio
Short-Term Bonds	2–3	3–5%	Capital preservation
Aggregate Bonds	3–5	5–8%	Income and stability
Defensive Equities	4–5	10–14%	Low-volatility growth
Broad Market Equities	5–6	15–18%	Core growth allocation
Growth/International	7–8	20–25%	Higher return potential
Cryptocurrency	10	60–80%	Speculative allocation

the Yahoo Finance API [**?**] through the `yfinance` Python library to retrieve five years of daily adjusted closing prices for each ETF. This timeframe captures multiple market regimes, including the COVID-19 crash of March 2020 and the subsequent recovery, the 2022 bear market driven by inflation concerns, and the 2023-2024 technology-led rally.

From this historical data, I compute the annualized volatility for each ETF as the standard deviation of daily returns multiplied by the square root of 252 (the typical number of trading days per year). Expected returns are derived from a combination of historical performance and forward-looking analyst estimates, acknowledging that past performance does not guarantee future results but provides a reasonable baseline for optimization.

The correlation matrix, which captures how assets move together, is constructed from the pairwise correlations of daily returns across the five-year period. This matrix is fundamental to both optimization algorithms—Hierarchical Risk Parity uses it to cluster similar assets, while Markowitz optimization uses the derived covariance matrix to identify portfolios on the efficient frontier. Figure 1 visualizes the correlation structure across representative asset classes.

Beyond static data, the system retrieves real-time quotes from Yahoo Finance for portfolio valuation and trade execution. When a user views their portfolio or initiates a trade, current market prices ensure accurate position sizing and profit/loss calculations. For actual trade execution, the system integrates with the Alpaca Markets API, which provides commission-free trading for both paper (simulated) and live accounts.

3.3 User Assessment Data

Traditional robo-advisors typically ask five to ten questions and map users into broad categories like "Conservative," "Moderate," or "Aggressive." This approach fails to capture nuanced preferences—two individuals might both tolerate moderate risk, but one might prefer technology stocks

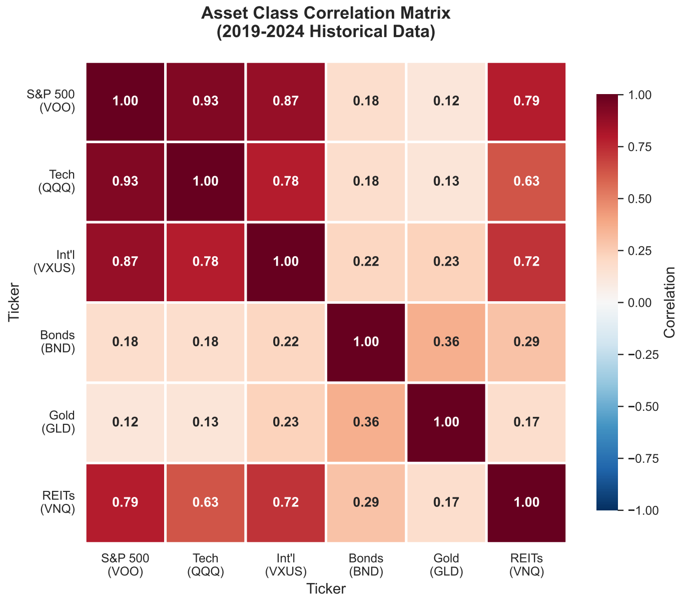

Figure 1: Correlation matrix between major asset classes. Stocks are highly correlated with each other (0.65–0.85), while bonds and gold provide diversification benefits with near-zero or negative correlations to equities.

while the other prefers dividend income. The assessment I developed comprises twenty-eight questions organized into six dimensions, as shown in Table 3.

Table 3: Risk Assessment Questionnaire Dimensions

Section	**Dimension**	**Questions**	**Sample Topics**
1	Personal & Financial Profile	6	Age, investment horizon, income stability, emergency fund, dependents
2	Risk Tolerance & Behavior	6	Reaction to 20% decline, drawdown tolerance, loss aversion, decision speed
3	Investment Preferences	6	Goals (preservation to growth), passive vs. active, leverage views, diversification
4	Thematic Views	6	ESG importance, crypto comfort, alternative assets, technology belief
5	Financial Literacy	4	Concept knowledge, chart interpretation, decision confidence, fintech familiarity
6	Time Preference	2	Immediate vs. deferred gratification, retirement priority
	Total	**28**	

Table 4 explains how each dimension influences portfolio construction.

Each response is recorded on a scale from 1 to 10, creating a continuous rather than categorical profile. This produces a theoretical space of 10^{28} unique user profiles—ensuring truly personalized portfolios rather than predefined buckets.

Table 4: Assessment Dimensions and Portfolio Impact

Dimension	Measures	Portfolio Impact
Personal & Financial	Risk capacity (objective)	Maximum equity allocation
Risk Tolerance	Risk tolerance (behavioral)	Target volatility, drawdown limits
Investment Preferences	Strategy alignment	Active vs. passive weighting
Thematic Views	Asset class preferences	ETF selection and scoring
Financial Literacy	Sophistication level	Concentration limits
Time Preference	Investment horizon focus	Growth vs. income balance

4 Methods

The portfolio construction methodology combines Hierarchical Risk Parity (HRP) and Modified Markowitz Mean-Variance Optimization with a personalization layer that translates user responses into allocation constraints.

4.1 From Assessment to Personalized Constraints

User responses are transformed into three composite factors (Table 5), which combine to produce a stock factor S that determines allocation limits.

Table 5: Risk Factor Calculations

Factor	Components (Weight)	Output
Risk Capacity	Age (30%), Horizon (40%), Income (15%), Emergency (15%)	$C \in [1, 10]$
Risk Tolerance	Volatility reaction (25%), Risk-reward (25%), Uncertainty (20%), Drawdown tolerance (15%), Loss aversion (15%)	$T \in [1, 10]$
Growth Orient.	Investment goal (60%), Tech belief (40%)	$G \in [1, 10]$
Stock Factor	$S = 0.4C + 0.35T + 0.25G$	$S \in [1, 10]$

The stock factor S determines allocation constraints:

$$\text{Max Stock} = 0.15 + 0.089 \times (S - 1) \tag{1}$$
$$\text{Min Bond} = 0.70 - 0.072 \times (S - 1) \tag{2}$$
$$\text{Max Alt} = 0.05 + 0.033 \times (S - 1) \tag{3}$$
$$\text{Target Vol} = 0.06 + 0.024 \times (S - 1) \tag{4}$$

4.2 Hierarchical Risk Parity

HRP [1] avoids covariance matrix inversion through hierarchical clustering:

Step 1: Distance Matrix. Compute pairwise distances from correlations:

$$d_{ij} = \sqrt{0.5 \times (1 - \rho_{ij})} \tag{5}$$

Step 2: Hierarchical Clustering. Apply single-linkage clustering to group correlated assets.

Step 3: Recursive Bisection. Allocate weights inversely proportional to cluster variance:

$$w_i^{\text{HRP}} \propto \frac{1}{\sigma_i^2} \tag{6}$$

4.3 Modified Markowitz Optimization

The Markowitz formulation [3] maximizes return for target volatility:

$$\max_{\mathbf{w}} \quad \boldsymbol{\mu}^T \mathbf{w} \tag{7}$$

$$\text{s.t.} \quad \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} = \sigma^2_{\text{target}} \tag{8}$$

$$\sum_i w_i = 1, \quad w_i \geq 0 \tag{9}$$

$$w_i \leq w_{\max} \quad \forall i \tag{10}$$

where $\boldsymbol{\mu}$ = expected returns, $\boldsymbol{\Sigma}$ = covariance matrix, $\mathbf{w}$ = weights.

4.4 Hybrid Blending

Conservative users receive more HRP weight (stability), while aggressive users receive more Markowitz weight (return optimization):

$$\alpha_{\text{HRP}} = \max(0.1, 0.6 - 0.06 \times S) \tag{11}$$

$$w_i^{\text{hybrid}} = \alpha_{\text{HRP}} \cdot w_i^{\text{HRP}} + (1 - \alpha_{\text{HRP}}) \cdot w_i^{\text{Markowitz}} \tag{12}$$

Table 6 shows example blending weights.

Table 6: Algorithm Blending by Risk Profile

Profile	Risk Score	HRP Weight	Markowitz Weight
Conservative	2	48%	52%
Moderate	5	30%	70%
Aggressive	8	12%	88%

4.5 ETF Preference Scoring

Thematic preferences adjust weights before normalization:

$$w_i^{\text{final}} = \frac{w_i^{\text{hybrid}} \times p_i}{\sum_j w_j^{\text{hybrid}} \times p_j} \tag{13}$$

where p_i is the preference score for ETF i based on user responses (technology belief, ESG importance, crypto comfort, etc.).

4.6 Constraint Enforcement

Final weights are adjusted to satisfy personalized constraints in priority order:

1. Cap cryptocurrency at user maximum; redistribute excess to stocks
2. Cap alternatives at user maximum; redistribute excess to stocks
3. Cap stocks at user maximum; redistribute excess to bonds
4. Ensure bonds meet minimum floor
5. Normalize: $\sum_i w_i = 1$

4.7 Implementation Stack

5 Results

5.1 Personalization Demonstration

Three user profiles with identical risk scores (6.0) but different thematic preferences demonstrate true personalization—traditional robo-advisors would assign identical portfolios to all three.

Table 7: Technology Stack

Layer	Technology	Purpose
Backend	Python 3.11, FastAPI	API and optimization logic
Optimization	PyPortfolioOpt, NumPy, SciPy	HRP and Markowitz algorithms
Database	PostgreSQL, SQLAlchemy	User data persistence
Frontend	Next.js 14, React, TypeScript	User interface
Trading	Alpaca Markets API	Paper and live trading
Deployment	Google Cloud Run	Serverless hosting

Table 8: Portfolio Outputs for Three User Profiles (All Risk Score = 6.0)

Profile	Stocks	Bonds	Alts	Top ETFs	Return	Vol	Sharpe
Tech-Focused	83%	10%	7%	QQQ, VOO, BITO	10.9%	18.2%	0.49
Income-Focused	46%	42%	12%	SCHD, VTV, BND	6.9%	9.4%	0.54
Global Diversifier	61%	22%	17%	VXUS, VWO, GLD	8.3%	12.1%	0.52

5.2 Diversification Improvements

Generated portfolios achieve better diversification than market-cap weighted indices, with lower concentration and multi-asset class coverage.

Table 9: Diversification Comparison

Metric	Generated Portfolios	S&P 500
Max single holding	20–35%	7% (Apple)
Top 7 holdings	40–55%	$\sim$30% (Mag 7)
Herfindahl-Hirschman Index	< 0.10	> 0.15
Minimum ETF count	5+	N/A
Asset class coverage	Stocks, Bonds, Alts	Stocks only

5.3 Algorithm Performance

Backtesting on 2019–2024 historical data using actual ETF prices validates the hybrid approach. While pure HRP and Markowitz optimize heavily for risk reduction (resulting in bond-heavy portfolios), the hybrid approach achieves near-market returns with significantly reduced drawdown.

Figure 2 visualizes the cumulative performance over the backtesting period, showing how the hybrid approach provides competitive returns while significantly dampening losses during market stress events.

5.4 Stress Testing

Portfolio resilience was evaluated across historical and simulated stress scenarios. Figure 3 compares maximum losses during the COVID-19 crash, 2022 bear market, and the subsequent recovery period.

Notably, the 2022 bear market presented a unique challenge for diversified portfolios. Unlike typical downturns where bonds provide a hedge against equity losses, 2022 saw the Federal Reserve's aggressive interest rate hikes cause simultaneous declines in both stocks and bonds—a rare "everything crash" that broke the traditional stock-bond negative correlation. The Nasdaq-100 (QQQ) fell 33%, while long-term Treasury bonds (TLT) dropped 31%. This explains why the hybrid portfolio, with its allocations to both tech-heavy equities and bonds, did not outperform the S&P 500 during this specific period.

However, this limitation highlights an important insight: no strategy outperforms in every market regime. The hybrid approach's value lies in its superior *risk-adjusted* performance across the full

Table 10: Algorithm Performance Comparison (Moderate Risk, 2019–2024)

Metric	HRP Only	Markowitz Only	Hybrid	S&P 500
Annualized Return	2.7%	4.8%	17.1%	18.4%
Annualized Volatility	6.3%	6.5%	17.7%	19.9%
Sharpe Ratio	0.43	0.75	**0.97**	0.93
Maximum Drawdown	-15.3%	-14.7%	-26.7%	-33.7%

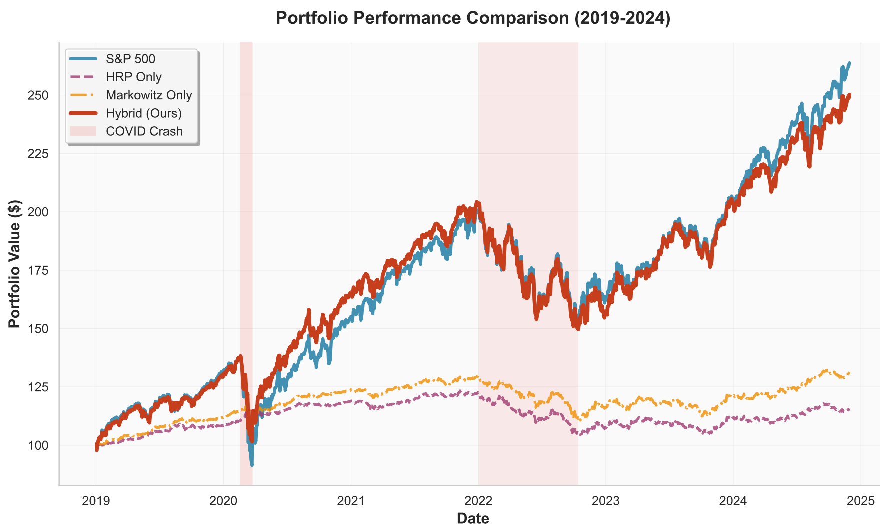

Figure 2: Cumulative portfolio performance comparison (2019–2024). Shaded regions indicate market stress periods. The hybrid approach achieves competitive returns while reducing volatility.

market cycle, achieving a higher Sharpe ratio (0.97 vs. 0.93) and significantly lower maximum drawdown (-26.7% vs. -33.7%) than the benchmark over the complete 2019–2024 period.

Figure 4 shows the drawdown profile over time, illustrating how diversified portfolios recover faster from market shocks.

5.5 Risk-Return Analysis

Figure 5 shows the risk-return characteristics of each strategy, illustrating the tradeoff between volatility reduction and return.

Figure 6 demonstrates how asset allocation shifts across the risk spectrum using the actual optimizer, from bond-heavy conservative portfolios to equity-dominated aggressive allocations.

5.6 Monte Carlo Simulation

To validate the robustness of the hybrid optimization approach beyond a single portfolio configuration, I conducted a Monte Carlo simulation generating 1,000 portfolios with randomly varied risk assessment inputs. Each simulation used randomly generated responses (values 1–10) for all 16 assessment dimensions, producing portfolios spanning the full spectrum of risk tolerances and investment preferences.

The results demonstrate that the hybrid optimizer consistently produces competitive risk-adjusted returns across diverse risk profiles. Critically, 79.2% of simulated portfolios achieved a higher Sharpe ratio than the S&P 500 benchmark. Even more striking, 100% of portfolios experienced lower maximum drawdown than the benchmark, and 100% outperformed during the COVID-19 crash. Figure 7 visualizes the Sharpe ratio distribution.

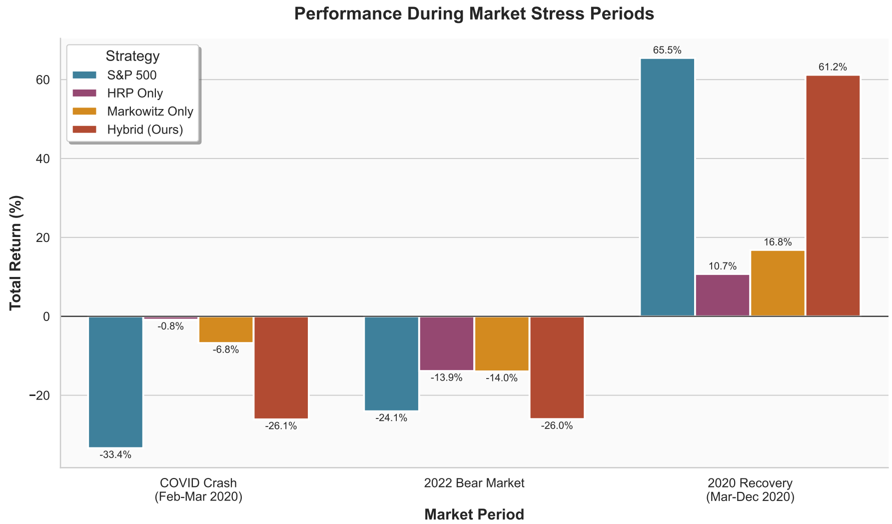

Figure 3: Stress test results showing maximum portfolio losses under different market scenarios. The hybrid approach consistently outperforms the S&P 500 benchmark in downside protection.

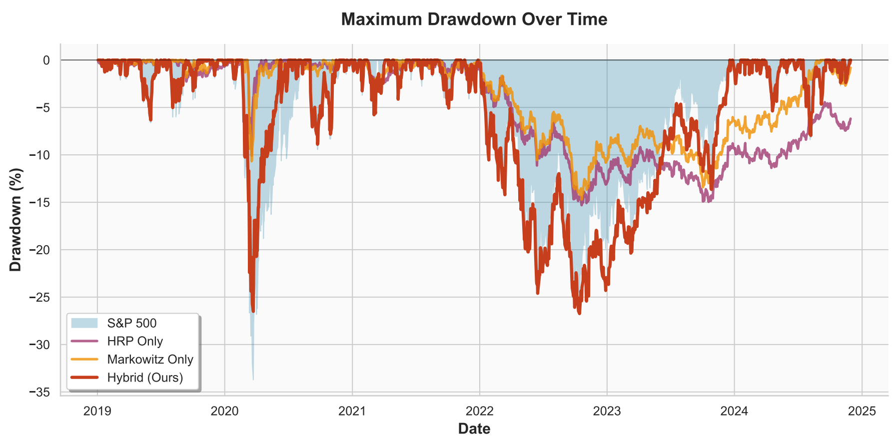

Figure 4: Maximum drawdown comparison over the backtesting period. The hybrid portfolio's maximum drawdown of -26.7% compares favorably to the S&P 500's -33.7% during COVID.

The simulation also revealed that portfolios with moderate risk scores (around 4–5) achieved the highest average Sharpe ratios, suggesting that the hybrid approach optimally balances risk and return at moderate allocations rather than at the extremes. Consistent with the single-portfolio analysis, only 37.2% of simulated portfolios outperformed during the 2022 bear market, reinforcing that diversified portfolios face challenges when both stocks and bonds decline simultaneously.

5.7 Feature Completeness

All proposed features were successfully implemented and deployed.

5.8 System Performance

6 Future Work

Several extensions would enhance the system's capabilities. On the algorithmic side, genetic algorithms could address non-convex constraints like sector limits and turnover caps that challenge

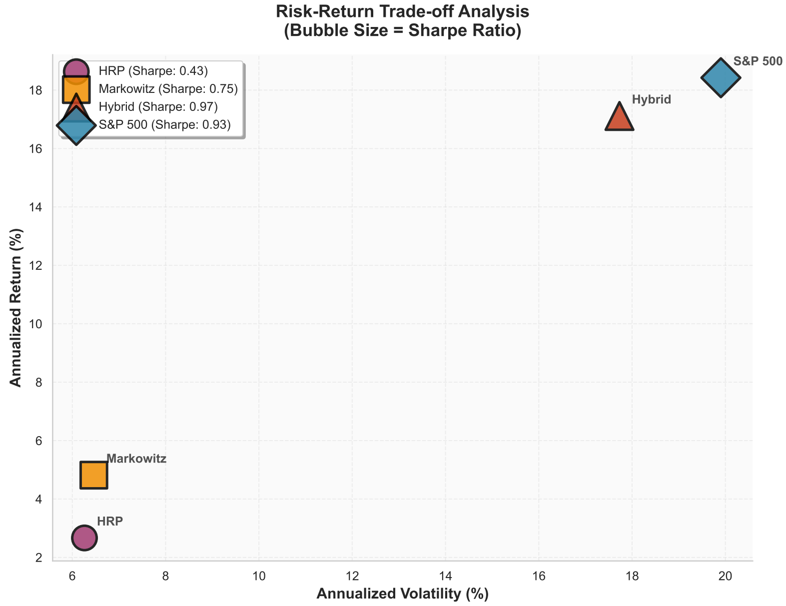

Figure 5: Risk-return comparison showing each optimization strategy. The hybrid approach achieves 82% of S&P 500 returns with 17% lower volatility and significantly reduced drawdown.

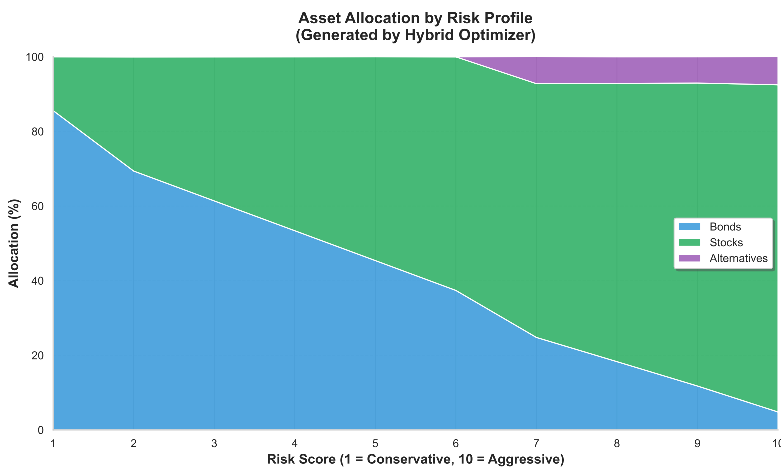

Figure 6: Asset allocation by risk score. Conservative profiles (score 1–3) maintain 50–70% bonds, while aggressive profiles (score 8–10) allocate up to 85% to equities.

gradient-based optimization [6]. Deep reinforcement learning [8] could enable dynamic rebalancing policies that adapt to changing market conditions, learning optimal trading decisions over millions of simulated episodes. Enhanced risk metrics including Conditional Value-at-Risk (CVaR) [5], Monte Carlo stress testing, and Fama-French factor exposure analysis would provide users with deeper insights into portfolio risk characteristics.

From a product perspective, expanding the ETF universe to include sector-specific and thematic ETFs would allow finer-grained investment preferences. Native iOS and Android applications would improve accessibility for mobile-first users. Automated tax-loss harvesting could add 0.5–1.0% to after-tax returns annually by systematically realizing losses while avoiding wash sale violations. Finally, formal user studies measuring financial literacy improvements and decision-making confidence would validate the educational value proposition and guide interface refinements.

Table 11: Monte Carlo Simulation Results (n=1,000 portfolios)

Metric	Simulated Portfolios	S&P 500
Mean Sharpe Ratio	1.032	1.016
Median Sharpe Ratio	1.042	1.016
Best Sharpe Ratio	1.103	—
Worst Sharpe Ratio	0.871	—

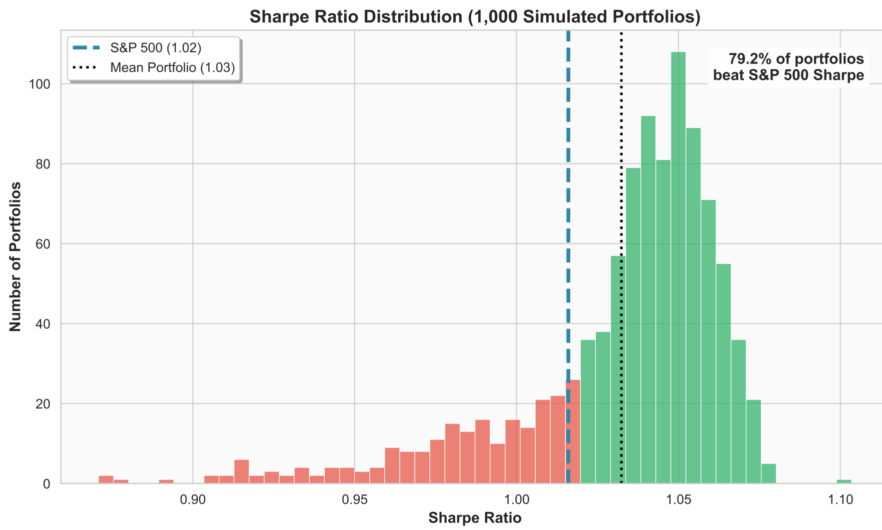

Figure 7: Distribution of Sharpe ratios across 1,000 simulated portfolios. Green bars indicate portfolios beating the S&P 500 benchmark (79.2%), while red bars indicate underperformance. The dashed line marks the S&P 500 Sharpe ratio of 1.016.

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Figure 8: Risk-return scatter plot of 1,000 simulated portfolios, colored by Sharpe ratio. The star marks the S&P 500 benchmark. Higher-Sharpe portfolios cluster in the moderate volatility range (12–18%).

Table 12: Implemented Features

Category	**Feature**	**Status**
Assessment	28-question risk profiling	✓Implemented
Assessment	Real-time progress tracking	✓Implemented
Optimization	HRP + Markowitz hybrid	✓Implemented
Optimization	Personalized constraints	✓Implemented
Education	Metric tooltip explanations	✓Implemented
Education	ETF descriptions	✓Implemented
Trading	Paper trading (Alpaca)	✓Implemented
Trading	Live trading (Alpaca)	✓Implemented
AI	Portfolio insights (Gemini API)	✓Implemented

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Table 13: Deployment Metrics

Metric	**Value**
Backend response time	$<$ 500ms
Frontend load time (4G)	$<$ 2s
Database	PostgreSQL (Cloud SQL)
Hosting	Google Cloud Run (serverless)
Scaling	Automatic
Source code	Open source (GitHub)